Computer Network (new), Ashar 25th - Spring 2025
Total marks: 100, Pass marks: 45
BoCE VI semester

Attempt all the questions.

1. a) What is computer network? Distinguish between OSI and TCP/P reference model. [2+5]

   b) What is ISDN? Explain about the ISDN architecture in detail with example. [2+6]

2. a) What is transmission media? Explain about any two-transmission media in detail. [2+5]

   b) What are the major functions of data link layer? Briefly explain CSMA/CD with a suitable diagram [3+5]

3. a) Discuss about the significance of routing in computer networking? Differentiate between link state routing and distance vector routing. [2+5]

   b) Explain the terms. (4x2)

      i) Dynamic Host Configuration Protocol ii) DNS server and queries

4. a) A 7-bit Hamming Code is received as 1011011. Assume even parity and state whether the received code is correct or wrong. If wrong locate the bit in error. [7]

   b) What is congestion control? Explain about leaky bucket algorithm of traffic shaping. (2+6)

   OR

   Briefly explain the various TCP congestion control policies.

5. a) Why port number is used in networking? Distinguish between TCP and UDP. How is TCP connection established? Explain. [2+2+4]

   OR

   Define SMTP, POP and IMAP with their port number.

   b) What are the drawbacks in IPV4? Which of these drawbacks do IPV6 solve? Explain [2+5]

6. a) What is network security? How can different types of firewalls enhance network security? (1+2+5)

   b) What is HTTP protocol? Explain RSA algorithm. [2+5]

7. Write short notes on (any two) (2*5)

   a) Recent Trends in Telecom Technologies: 2G/3G/4G/5G.
   b) Switch
   c) NGN
   d) NAT

Computer Network, Magh 29th - Fall 2024
Total marks: 100, Pass marks: 45
BSE V semester

Attempt all the questions.

1. a) What is computer network? Distinguish between OSI and TCP/P reference model. [2+5]

   b) What is ISDN? Explain about the ISDN architecture in detail with example. [2+6]

2. a) What is transmission media? Explain about any two-transmission media in detail. [2+5]

   b) What are the major functions of data link layer? Briefly explain CSMA/CD with a suitable diagram [3+5]

3. a) Discuss about the significance of routing in computer networking? Differentiate between link state routing and distance vector routing. [2+5]

   b) Explain the terms. (4x2)

   i) Simple Mail Transfer Protocol ii) Domain Name System

   7654321

4. a) A 7-bit Hamming Code is received as 1011011. Assume even parity and state whether the received code is correct or wrong. If wrong locate the bit in error. [7]

   b) What is congestion control? Explain about leaky bucket algorithm of traffic shaping. (2+6)

   OR

   Briefly explain the various TCP congestion control policies.

5. a) Why port number is used in networking? Distinguish between TCP and UDP. How is TCP connection established? Explain. [2+2+4]

   25 110 143

   OR

   Define DHCP, SMTP, POP and IMAP with their port number.

   b) What are the drawbacks in IPV4? Which of these drawbacks do IPV6 solve? Explain [2+5]

6. a) What is network security? How can different types of firewalls enhance network security? (1+2+5)

   b) What is HTTP protocol? With an example explain how a request initiated by a HTTP client is served by a HTTP server [2+5]

7. Write short notes on (any two) (2*5)

   a) Recent Trends in Telecom Technologies: 2G/3G/4G/5G.
   b) Switch
   c) NGN
   d) NAT

a –
p –
s –
T –
Net

N → Internet
D → link
Ph → sendin

NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Spring 2025

Level: Bachelor

Program: BE CE_IT_M_D

Course: : **Computer Networks**

Semester: VI

Year : 2025

Full Marks: 100

Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. List out the devices used in computer networking and differentiate peer to peer and client-server architecture model. [8]

   b. Explain each layer of TCP/IP in brief mentioning at least two protocols and their core functions of each layer. [7]

2. a. Why optical fiber are best among the transmission media? Give reasons. How is circuit switching different from packet switching? [8]

   b. What factors determine the size of window? Explain the working principle of GO BACK N ARQ process of error handling. [7]

3. a. Illustrate with the help of an example the checksum method at both receivers and transmitters end. [8]

   b. Explain an algorithm for CSMA/CD with the help of flowchart. [7]

4. a. Given an IP address of 197.167.22. 0 / 24 create subnetworks for four departments with required hosts 6, 61, 13 and 9 respectively. Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. [8]

   b. What is routing? Mention its types. Explain with an example the distance vector routing protocol. [7]

5. a. List out advantages of IPv6 Header format over IPv4. Mention three types of transition strategies from IP4v to IPv6. [8]

b. Compare TCP and UDP header format and compare them. [7]

6. a. What is congestion? How TCP manages congestion? Explain the concept of leaky bucket algorithm for congestion control. [8]

OR

Explain the uses of NAT (Network Address Translation), ARP (Address Resolution Protocol) and RARP (Reverse Address Resolution Protocol).

b. What is Network Management System? Write down the steps to create a public private key pair with the help of RSA algorithm. [7]

OR

Explain the process of cryptography and explain the key exchange procedure in symmetric cryptography. List some of the malicious things intruders can do.

7. Write short Notes on (Any Two) [5 X 2 = 10]

a) ISDN
b) VPN and Firewall
c) DHCP.

NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Fall 2024

Level: Bachelor
Programme: BE_SE_M
Course: : Computer Networks
Semester: V

Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

1. a. What are the merits and demerits of computer networking? Differentiate between peer to peer and client-server architecture model. [8]

   b. Explain each layer of TCP/IP in brief mentioning at least two protocols and their core functions of each layer. [7]

2. a. Describe different types of transmission media used to exchange data between end devices with examples. How is circuit switching different from packet switching? [8]

   b. How does the size of window decided? Explain the working principle of GO BANCK N ARQ process of error handling. [7]

3. a. Illustrate with the help of an example the cyclic redundancy check method to detect error at receiver's end. [8]

   b. Explain an algorithm for CSMA/CD with the help of flowchart. What do you understand by 1 persistent, non-persistent and p-persistent method of medium access? [7]

4. a. Imagine yourself as a skillful network administrator in setting up computer networks. Given an IP address of 192.168.1. 0/24, you are assigned the responsibility to create four subnetworks for four departments namely Exam, Lab, Library and Accounts

that require 7, 49, 14 and 9 hosts respectively. Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. [8]

b. What is unicasting, multicasting and broadcasting? Explain with an example the distance vector routing protocol. [7]

5. a. Compare IPv4 Header format with IPv6 Headers. Mention three types of transition strategies from IP4v to IPv6. [8]

b. Compare TCP and UDP header format and compare them. [7]

6. a. What is congestion? How TCP manages congestion? Explain the concept of leaky bucket algorithm for congestion control. [5]

OR

How NAT (Network Address Translation) does performs its task?

b. Write down the steps to create a public private key pair with the help of RSA algorithm. [5]

OR

Explain the process of cryptography and its importance to maintain security triad.

c. What is Network Management System? Describe its major components. List some of the malicious things intruders can do. [5]

7. Write short Notes on (Any Two) [5 X 2 = 10]

a) ISDN
b) VPN and Firewall
c) Webserver and DNS.

Nepal College of Information Technology
Unit Test 2025-Spring

Level: Bachelor | Year : 2025
Programme: BE_CE_M_D | Full Marks: 70
Course: : **Computer Networks** | Pass Marks: 30
Semester: VI | Time : 2 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. What is computer networking? Differentiate between peer to peer and client-server architecture model. [8]

   b. Explain each layer of TCP/IP in brief and compare it with OSI reference model. [7]

2. a. Describe different types of transmission media used to exchange data between end devices with examples. How is circuit switching different from packet switching? [8]

   b. How does the size of window decided? Explain the working principle of GO BANCK N ARQ process of error handling. [7]

3. a. Illustrate with the help of an example the cyclic redundancy check method to detect error at receiver's end. [8]

   b. Explain an algorithm for CSMA/CD with the help of flowchart. What do you understand by 1 persistent, non-persistent and p-persistent method of medium access? [7]

4. a. Imagine yourself as a skillful network administrator in setting up computer networks. Given an IP address of 192.100.1. 0/24, you are assigned the responsibility to create four subnetworks for four branches of a finance company at different locations with following hosts requirements 7, 13, 55 and 100 respectively.

Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. [8]

b. What is unicasting, multicasting and broadcasting? Explain with an example the distance vector routing protocol. [7]

7. Write short Notes on (Any Two) [5 X 2 = 10]

a) ISDN
b) Frame Relay
c) Virtual Circuit Packet Switching.

NEPAL ENGINEERING COLLEGE

Level: Bachelor Semester: Spring Year: 2025
Program: BE Full Marks: 100
Course: Computer Network_ New Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1 a) Define computer networks and discuss their merits and demerits. Compare PAN, LAN, MAN, and WAN. 7

b) Explain the OSI model with a focus on the functions of the Transport and Data Link layers. 8
OR
Describe the TCP/IP model and compare it with the OSI model.

2 a) Discuss guided and unguided transmission media with examples. Explain the concept of circuit switching and packet switching. 7

b) Explain ISDN signaling and its architecture. 8

3 a) Describe the LLC and MAC sublayers of the Data Link layer. Explain framing and flow control mechanisms. 7

b) Discuss CSMA/CD and its role in Ethernet networks. 8

4 a) Explain IPv4 addressing, subnetting, and CIDR. Compare IPv4 and IPv6 headers. 7
OR
Create 32 Subnet of given IP address 172.30.0.0/16 with new subnet mask and find out the network ID, Wildcard Mask, Usable Host IP & Broadcast Mask.

b) Describe distance vector and link-state routing algorithms. Provide examples of RIP and OSPF. 8

5 a) Explain the services and features of TCP and UDP. Discuss the concept of socket programming. 7

b) Discuss congestion control mechanisms (open loop and closed loop) and traffic shaping algorithms (Leaky Bucket, Token Bucket). 8
OR
Compare TCP and UDP segment headers. (Unit VI)

6 a) Discuss symmetric key (DES) and asymmetric key (RSA) cryptography. 7

b) Describe the role of firewalls & its types and digital certificates in network security. 8

7 Write Short notes (Any Two) 2*5

a) NGN

b) E-Mail Services

c) ATM

Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Assessment (Spring, 2025)

Level: Bachelor | Full Marks: 100
Programme: Computer Engineering | Pass Marks: 45
Year/Part: III/II | Time: 3 hrs

Subject: - Computer Network

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all questions*

1. a). Define computer network along with its architecture? Explain different kinds of services provided by the network technologies in real life scenario? [4+4]

   b). What is protocol? Why do we need layered protocol architecture? Explain each layer of ISO OSI layered architecture with neat diagram [7]

2. a). Define transmission media? Explain about different kinds of guided media with suitable examples [7]

   b). What do you mean by CSMA/CD and CSMA/CA? Design an algorithm and flow chart for CSMA/CD technology [8]

3. a). Write any two differences between IPV4 and IPV6? Explain the frame format of IPV4 with suitable diagram? [2+5]

   b). What is the function of Data link layer? Explain about framing in detail [2+6]

4. a). Our MBCOE has three departments: Computer, Civil and Architecture with 45, 65, and 22 hosts. Explain how will you design network for MBCOE. Let you are provided IP address 10.12.100.0/24. Find network address, broadcast

address, subnet mask, wildcard mask and usable IP pool for each subnet [8]

b). What is adaptive and non- adaptive routing? Compare classful and classless routing with suitable examples [7]

5. a) What do you mean by socket address.? Explain the concept of TCP and UDP socket programming with suitable diagram [8]

b). What are the reasons for the occurrence of congestion? Explain the open loop solutions of congestion? [7]

6. a). Explain different types of flow control mechanism with suitable examples? [8]

OR

a). List out different type of error detection mechanism. Do you know any method that is used for the error detection and correction as well? Describe with suitable examples [8]

b) What is cryptography? Explain the operation of DES or RSA algorithm with suitable example [7]

7. Write short notes on: [Any Two] [5+5]

a. IPsec
b. Firewall
c. DNS server

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre University Examinations

Level: Bachelor | Semester: VI_Spring | Year : 2025

Program: B.E. Computer | F.M. : 100

Course: **Computer Network** | P.M. : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network along with its applications. Explain about Network Topology? 7

   b) Define Protocol. Why do we need layered protocol architecture? Discuss each layer of TCP/IP protocol architecture in detail. 8

2. a) Justify the implications of optical fiber in present day to enable global communication with a suitable example. 8

   b) What is Error Detection? Show how CRC technique will help to detect the error with suitable example. 7

3. a) Design an algorithm for CSMA/CD with a suitable example. 8

   b) Why do we require a routing table? Explain distance vector routing algorithm with suitable diagram. 7

4. a) *Pokhara University is a Member of Regional Internet Registry*. It has acquired an IPv4 pool of (105.98.0.0/22). From the acquired pool, it is required to be subnetted and distributed across four schools. Among four schools, one school need to be subdivided into two different departments. Design and provide a complete IP Address plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask for every subnet. 8

   b) What is port addressing? Explain the working mechanism of TCP Socket Programming. 7

5. a) Explain block diagram of UDP socket. Explain DHCP operation. 7

   b) Discuss the role of SMTP. Explain the operation of DNS in corporate networks. 8

6. a) Explain the working principle of HTTP. Explain IMAP protocol. 8

   b) Explain Diffie Hellman Key Exchange Protocol with a suitable Example. 7

7. Write short notes on: (Any two) 2x5=10

a) Frame Relay

b) TCP three-way Handshake

c) Firewall and its types

CamScanner

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre University Examination

Level: Bachelor
Semester – Spring VI/VII
Year : 2025
Programme: BE Computer
Full Marks : 100
Course: **Computer Network**
Pass Marks : 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Discuss the merits and demerits of computer network. Compare star, bus, ring and mesh topology. [7]
   b) Explain the working principle of different types of network devices: hub, switch, bridge, and router. [8]
2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. [7]
   b) What is error control. Explain CRC method of error detection with a suitable example. [8]
3. a) What are intermediate devices in a network? Describe how fiber optics cable provides faster, reliable and efficient means for data transmission. [7]
   b) What are the components of IPV4 frame format? Differentiate between IPV4 and IPV6. [4+4]
4. a) NAST College is planning to design 8 different department for BE Computer, BCA, BBA, BE Civil, Exam, Research, Account and library. Each department consists of 30 computers with IP address192.168.218.0. Explain the process how will you allocate the IP address and subnet for each of the departments using above IP. [8]
   b) Discuss about Interior and Exterior Routing protocols. Explain any one routing protocol. [7]
5. a) What is congestion in a network. Explain about Leaky bucket algorithm with suitable diagram and algorithm. [8]
   b. List out and explain TCP client/server socket flow with suitable diagram. [7]

6. a) What is computer network. Explain different types of computer networks with suitable block diagram [2+6]

   b. Define IP addressing. Explain different classes of IP addressing with default network ID, Host ID, IP address Range and maximum capacity of host devices. [7]

7. Write short notes on: (**Any two**) [2x5=10]
   a) Frame Relay
   b) Recent Trends in Telecom Technologies: 2G/3G/4G/5G.
   c) Circuit Switching
   d) CSMA/CD

Carrier Sensing Multiple Access / Collision Detection,

PAN, WAN, LAN, GAN, CAN*, CAN, MAN,

COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2082/04/04			
Level	BE	Full Marks	70
Programme	BEIT, BCE	Time	
Semester	VI	2 hrs	

Subject: - Computer Network

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt **70 marks** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. a) Explain the OSI reference model with its layers and their functions. Compare it with the TCP/IP model. [8]
   b) List and briefly describe the different types of network topologies. [7]

2. a) Discuss the error detection and correction mechanisms used in the Data Link Layer, with examples. [8]
   b) What is Flow control? Explain Stop-and-Wait flow control mechanism. [7]

3. a) Explain IPv4 header format. [8]
   b) Perform sub-netting for the IP address 192.168.10.0/24 to create 4 subnets. Show the subnet mask, network IDs, and broadcast addresses for each subnet. [7]

4. a) Compare TCP and UDP protocols in terms of their features, headers, and applications. [8]
   b) What is socket programming? Explain the difference between TCP and UDP sockets. [7]

5. a) Describe the working principles of congestion control algorithms (Leaky Bucket and Token Bucket) in computer networks. [8]
   b) Explain in detail the TCP connection establishment with proper diagrams. [7]

6. a) Explain the role of DNS in the Application Layer. Describe recursive and iterative DNS resolution. [8]
   b) What are the differences between symmetric and asymmetric key cryptography? Provide examples of each. [7]

7. Write short notes on: (**any two**) [2*5=10]
   a) **CSMA/CD**
   b) **Wireshark**
   c) **Cisco Packet Tracer**

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE
INTERNAL ASSESMENT

Level: Bachelor Semester: 5th Fall 2024 Year : 2025
Programme: BE Software Full Marks: 100
Course: Computer Network Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) You have been tasked with designing your University network infrastructure. Suggest a University-useful network solution that combines the newest hardware and software trends. Make the required assumptions and use logical arguments to support your recommendation. 8

   b) Describe the merits and demerits of Computer Network. Differentiate Client Server and Peer to Peer Network Models. 7

OR

Explain the term bandwidth, throughput and latency. Describe the design issue of the different layers.

2. a) Explain message switching, circuit switching and packet switching in detail with appropriate figures. 8

   b) What are the different error detection mechanism. Explain CRC with one example. 7

OR

7-bits even parity Hamming code is received as 1100010. Find the correct code that was transmitted.

3. a) What are the different Flow control mechnisam. Explain in detail. 8

OR

Explain IPV6 header format. Compare it with IPV4 header format.

b) Explain Distance Vector Routing and Link State Routing with appropriate examples. 7

c) Design a network for GCES having 5 departments having hosts 5,16,23,27 such that IP address wastage in each department is minimum. Find out subnet mask, network address, broadcast address and wildcard mask and usable host range. The given IP is 10.10.10.0/24. 8

4. a) Explain the concept of socket programming in TCP and UDP socket. 8

   b) Explain Leaky bucket and Token bucket algorithm for congestion control with suitable diagram. 7

OR

What are the different TCP and UDP services ? Explain TCP handshaking.

5. a) Prepare the table of difference between RIP,OSPF,BGP. What are the special characteristic of BGP ? 7

   b) What do you mean by DHCP and DNS ? How are they implemented in Networking ? 8

6. a) Explain RSA cryptography algorithm with example. What do you mean by Digital Certificate ? 7

   b) Explain Deffie Hellman key exchange. Explain Kerberos. 5

7. Write short notes on the following **(Any Two)**: (2.5+2.5) 5

   a) CSMA/CD
   b) Proxy Server
   c) Framing
   d) LLC and MAC sublayer

Lumbini Engineering College
(Final Internal Exam)

Level: Bachelor
Programme: BE Computer 6th
Course: Computer Networks

Year : 2082
Full Marks : 100
Pass Marks : 45
Time : 3 hrs

Candidates are required to give their answer in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions.

1. a) What is transmission impairments? State and explain in brief different factors of transmission impairments. 7
   b) State pros and cons of Computer Network. Explain different Computer network models. 8
2. a) What is switching? Make comparisons between circuit switching and packet switching. 8
   b) Why Computer network is decided to split into layers? State and explain in brief about OSI reference model. 7
3. a) Classify the transmission media. Make comparison between base band and broad band cable. 7
   b) What is framing and flow control? Explain in brief about Leaky bucket algorithm. 8
4. a) Describe TCP handshaking mechanism with diagram. 8
   b) What is unicasting and multicasting. Explain in brief about IPV4 frame format. 7
5. a) A company uses the IP address 197.70.64.10/9.What does it mean? Explain in brief about all the factors related with it. 8
   7

   b) What is CRC? Explain in brief about HDLC protocol header format. 8
6. a) What is choking? Classify the routing algorithms. State and explain in brief about congestion control algorithm used in routing. 7
   b) What are the necessity of cryptography in communication? Explain in brief about public key algorithm.
7. Write short notes on any two: 2×5
   a) DNS
   b) DHCP
   c) Micro Wave and Satellite

UNITED TECHNICAL COLLEGE

Level: Bachelor | Semester – Spring | Year : 2025
Programme: BE | | Full Marks: 100
Course: Computer Network | | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1.(a) Define a computer network. Explain the types of networks: PAN, LAN, CAN, MAN, WAN, and GAN on the basis of their coverage, use cases, and real-world examples. 7

(b) Compare the Client-Server and Peer-to-Peer networking models on the basis of architecture, scalability, fault tolerance, and typical applications. 8

2.(a) Describe the OSI seven-layer model on the basis of layer functions, services provided, and example protocols at each layer. 7

(b) A network link has a bandwidth of 200 Mbps and a propagation delay of 15 ms. Calculate the bandwidth-delay product and explain its role in network performance. 8

3.(a) Explain Go-Back-N and Selective Repeat ARQ protocols on the basis of their mechanism, efficiency, retransmission process, and frame buffering. Support your answer with appropriate diagrams. 7

(b) Explain the process of IPv4 subnetting with a complete numerical example. Also define CIDR and VLSM, and describe their roles in IP address optimization. 8

4. (a) For an 8-bit data sequence 10110110, generate the Hamming code. If the received sequence is 101101101011, detect and correct any single-bit error. 7

(b) Explain TCP Congestion Control on the basis of its four phases: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery. Support your answer with a graph illustrating congestion window growth. 8

5. (a) Explain traffic shaping algorithms using Token Bucket and Leaky Bucket, on the basis of their operation, burst control, and data flow regulation. Include a diagram and example scenario for each. 7

(b) What is "socket programming "and explain its relevance in network application development. *What is a "Next Generation Network (NGN)"? Highlight its key characteristics.* 8

6.(a) Describe how the Domain Name System (DNS) works. On the basis of resolution methods, differentiate between recursive and iterative DNS queries, with examples of query flow. 7

(b) Explain the principles of symmetric encryption (DES) and asymmetric encryption (RSA) on the basis of key usage, speed, and security level. Include real-world use cases of each. 8

7. Short Notes: (Attempt any two)

a. CSMA/CA

b. ISDN

c. VPN

d. Frame Relay

Pokhara University

Everest Engineering College

Final Internal Assessment

Spring- 2025

Level: Bachelor

F.M. 100

Program: BE CMP(6th Semester)

P.M. 45

Faculty: Science & Technology

Time: 3hrs

Section:

Subject: Computer Network

Attempt all the questions.

1	a)	Explain how client server architecture is more secure than peer to peer networks?	7
	b)	Discuss the Seven Layers of OSI protocol stack. Also compare TCP/IP and OSI layers.	8
2	a)	Discuss how packet switching is different from circuit switching	7
	b)	Differentiate between error detection and error correction. Describe how Hamming code works for data of 7 bits. Show that it can detect single bit error using an example.	8
3	a)	Explain IPV4 header format. Differntiate between IPv4 and IPV6	7
	b)	The APNIC Pool for Pokhara University (103.16.32.0/22) is to be divided into network of 7 different schools. Among seven schools three school need to be subdivided into two different departments. Provide a complete IP address plan which include network address, broadcast address, usable IP pool, subnet masks and wildcard masks	8
4	a)	Why congestion occurs in the network? Explain the type of closed loop congestion.	8
	b)	Define DHCP. Explain the iterative and recursive DNS query for name resolution.	7

5 a) What is Adaptive and Non Adaptive Routing? State and explain Link state Routing with example. 7

b) Explain the working Principle of RSA algorithm. If N=119, public key E=5, and private key D=77, then demonstrate how to end the character plain text PT=6 using RSA 8

6 a) Discuss VPN and its types. 7

b) What are the different guided media explain in brief 8

7 Write short notes on: (Any two) **2*5=10**

a) Proxy Server
b) Frame Relay
c) SMTP

Best Wishes